LSHADE — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Nota sobre este algoritmo

L-SHADE con su propia poblacion (reduccion lineal), no N=80. Presupuesto (T+1)*N = 5040 / 50 080 / 500 080, igual que los clasicos.

Resumen — posicion entre los 10 algoritmos (rango por media)

dimension	presupuesto	rango medio de Friedman	posicion de 10	funciones donde es el mejor
D10	5k FES	2.76	3º	2 de 71
D10	50k FES	2.53	2º	27 de 71
D10	500k FES	2.85	2º	23 de 71
D50	5k FES	5.49	6º	1 de 71
D50	50k FES	2.43	2º	11 de 71
D50	500k FES	2.16	2º	33 de 71

LSHADE — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	2.33e+03	2.83e+05	1.99e+04	3.01e+04	4.08e+04	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22014	2014	10	7.39e+03	5.11e+05	8.54e+04	1.16e+05	1.11e+05	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F32014	2014	10	2.91e+00	3.87e+02	3.69e+01	6.90e+01	8.16e+01	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F42014	2014	10	6.00e-01	3.57e+01	3.48e+01	3.35e+01	6.61e+00	0.00e+00	3.48e+01	3.48e+01	3.21e+01	9.44e+00	0.00e+00	3.48e+01	3.48e+01	3.08e+01	1.11e+01
F52014	2014	10	2.01e+01	2.06e+01	2.03e+01	2.03e+01	1.01e-01	1.74e+00	2.01e+01	2.00e+01	1.65e+01	6.88e+00	5.68e-14	2.00e+01	1.16e+00	6.36e+00	8.92e+00
F62014	2014	10	8.67e-01	5.34e+00	3.17e+00	3.13e+00	1.14e+00	0.00e+00	2.00e-03	0.00e+00	4.06e-05	2.80e-04	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F72014	2014	10	2.78e-01	8.83e-01	6.87e-01	6.50e-01	1.62e-01	0.00e+00	2.22e-02	4.06e-11	1.47e-03	4.51e-03	0.00e+00	3.90e-08	0.00e+00	7.64e-10	5.46e-09
F82014	2014	10	4.28e+00	2.30e+01	9.31e+00	1.04e+01	4.04e+00	0.00e+00	1.14e-13	0.00e+00	8.92e-15	3.09e-14	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F92014	2014	10	9.79e+00	3.56e+01	2.18e+01	2.22e+01	6.32e+00	1.99e+00	5.98e+00	3.01e+00	3.27e+00	9.88e-01	1.46e-11	3.98e+00	2.98e+00	2.46e+00	8.75e-01
F102014	2014	10	5.34e+01	5.16e+02	1.73e+02	2.04e+02	1.04e+02	9.09e-13	1.25e-01	9.09e-13	2.08e-02	3.23e-02	0.00e+00	9.09e-13	0.00e+00	3.03e-13	4.33e-13
F112014	2014	10	4.76e+02	1.37e+03	9.55e+02	9.54e+02	1.90e+02	7.17e+00	2.40e+02	2.80e+01	5.63e+01	5.63e+01	3.76e-01	1.22e+02	1.18e+01	1.70e+01	1.82e+01
F122014	2014	10	4.24e-01	2.16e+00	9.51e-01	9.55e-01	2.79e-01	7.62e-02	1.74e-01	1.14e-01	1.17e-01	2.47e-02	1.29e-02	3.88e-02	2.26e-02	2.35e-02	6.10e-03
F132014	2014	10	1.40e-01	4.20e-01	2.69e-01	2.75e-01	6.61e-02	1.70e-02	1.13e-01	6.83e-02	6.81e-02	2.12e-02	1.27e-02	6.30e-02	3.52e-02	3.55e-02	9.97e-03
F142014	2014	10	1.28e-01	4.37e-01	3.02e-01	2.95e-01	7.37e-02	2.61e-02	1.69e-01	8.29e-02	8.52e-02	2.74e-02	1.97e-02	1.03e-01	5.00e-02	5.26e-02	1.98e-02
F152014	2014	10	1.45e+00	4.25e+00	3.28e+00	3.10e+00	7.17e-01	2.32e-01	6.82e-01	4.60e-01	4.56e-01	8.47e-02	2.15e-01	3.92e-01	3.00e-01	2.99e-01	4.56e-02
F162014	2014	10	2.49e+00	3.88e+00	3.24e+00	3.21e+00	3.50e-01	6.80e-01	2.20e+00	1.56e+00	1.50e+00	3.47e-01	3.97e-01	1.60e+00	8.91e-01	9.51e-01	2.82e-01
F172014	2014	10	1.25e+02	9.20e+03	1.06e+03	1.40e+03	1.41e+03	0.00e+00	2.08e-01	2.05e-12	7.75e-02	1.02e-01	0.00e+00	3.37e+02	0.00e+00	1.32e+01	6.60e+01
F182014	2014	10	4.50e+01	3.14e+03	3.12e+02	4.68e+02	5.01e+02	5.16e-02	1.42e+00	3.99e-01	3.97e-01	1.83e-01	6.60e-03	5.00e-01	1.38e-01	2.00e-01	1.77e-01
F192014	2014	10	4.36e+00	2.29e+01	8.33e+00	1.02e+01	4.78e+00	9.74e-01	2.85e+00	1.95e+00	1.91e+00	4.71e-01	7.60e-02	1.99e+00	6.54e-01	6.80e-01	4.42e-01
F202014	2014	10	2.64e+01	2.69e+04	4.09e+03	5.53e+03	5.58e+03	8.43e-02	9.63e-01	5.72e-01	5.28e-01	1.96e-01	9.60e-04	4.97e-01	1.57e-01	1.97e-01	1.50e-01
F212014	2014	10	2.38e+01	6.76e+03	3.93e+02	6.41e+02	1.01e+03	1.44e-03	1.13e+00	4.72e-01	4.66e-01	3.14e-01	2.98e-07	8.40e-01	3.12e-01	2.51e-01	2.58e-01
F222014	2014	10	2.75e+01	2.52e+02	3.88e+01	4.94e+01	4.25e+01	5.87e-01	2.38e+02	2.07e+01	2.08e+01	3.19e+01	1.68e-01	2.06e+01	5.51e-01	2.94e+00	6.46e+00
F232014	2014	10	2.42e+02	2.42e+02	2.42e+02	2.42e+02	1.11e-02	2.37e+02	2.42e+02	2.42e+02	2.42e+02	7.56e-01	2.37e+02	2.42e+02	2.42e+02	2.42e+02	1.28e+00
F242014	2014	10	1.19e+02	1.41e+02	1.32e+02	1.32e+02	5.31e+00	1.00e+02	1.13e+02	1.08e+02	1.08e+02	2.06e+00	1.00e+02	1.10e+02	1.07e+02	1.07e+02	1.62e+00
F252014	2014	10	1.86e+02	2.01e+02	2.00e+02	2.00e+02	2.67e+00	2.00e+02	2.00e+02	2.00e+02	2.00e+02	0.00e+00	1.34e+02	2.00e+02	2.00e+02	1.96e+02	1.58e+01
F262014	2014	10	1.00e+02	1.00e+02	1.00e+02	1.00e+02	7.15e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	1.83e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	8.73e-03
F272014	2014	10	5.95e+00	4.01e+02	4.00e+02	2.76e+02	1.75e+02	1.91e+00	4.00e+02	4.00e+02	3.05e+02	1.70e+02	1.04e+00	4.00e+02	4.00e+02	2.83e+02	1.83e+02
F282014	2014	10	3.59e+02	4.90e+02	3.73e+02	3.97e+02	4.61e+01	3.69e+02	4.81e+02	3.69e+02	3.89e+02	3.94e+01	3.57e+02	4.81e+02	3.69e+02	3.89e+02	4.16e+01
F292014	2014	10	4.50e+06	5.80e+06	5.31e+06	5.00e+06	4.08e+05	4.50e+06	5.31e+06	5.31e+06	5.06e+06	3.73e+05	4.50e+06	5.80e+06	5.31e+06	5.04e+06	4.15e+05
F302014	2014	10	2.76e+03	1.46e+05	4.97e+04	4.37e+04	3.14e+04	2.31e+03	4.90e+04	2.31e+03	1.84e+04	2.22e+04	2.31e+03	4.90e+04	2.31e+03	2.42e+04	2.35e+04
F12017	2017	10	8.50e+03	8.25e+05	1.06e+05	1.82e+05	1.81e+05	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22017	2017	10	6.39e-02	2.78e+01	1.67e+00	3.54e+00	4.98e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F32017	2017	10	4.23e+00	9.21e+00	8.64e+00	7.84e+00	1.71e+00	0.00e+00	3.99e+00	3.99e+00	2.66e+00	1.90e+00	0.00e+00	3.99e+00	3.99e+00	2.74e+00	1.87e+00
F42017	2017	10	2.15e+01	6.06e+01	4.19e+01	4.20e+01	9.18e+00	4.86e-03	3.99e+00	2.03e+00	2.39e+00	7.95e-01	2.64e-05	2.98e+00	1.99e+00	1.72e+00	6.62e-01
F52017	2017	10	2.76e-06	6.09e-04	7.11e-05	1.13e-04	1.25e-04	1.02e-11	5.56e-09	5.26e-10	7.63e-10	9.42e-10	0.00e+00	1.99e-12	5.68e-14	3.19e-13	4.68e-13
F62017	2017	10	4.58e+01	1.04e+02	7.42e+01	7.49e+01	1.22e+01	1.11e+01	1.45e+01	1.23e+01	1.24e+01	8.00e-01	1.10e+01	1.31e+01	1.17e+01	1.17e+01	4.90e-01
F72017	2017	10	1.91e+01	4.95e+01	3.00e+01	3.04e+01	7.11e+00	2.03e+00	1.20e+01	1.00e+01	9.11e+00	2.01e+00	2.00e+00	1.00e+01	6.00e+00	6.43e+00	1.71e+00
F82017	2017	10	8.64e-05	8.96e-02	1.12e-03	3.34e-03	1.24e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F92017	2017	10	3.42e+02	1.28e+03	7.47e+02	7.74e+02	2.08e+02	3.42e+00	4.09e+01	1.28e+01	1.50e+01	9.12e+00	2.50e-01	1.51e+01	6.89e+00	6.39e+00	3.55e+00
F102017	2017	10	2.55e+01	5.38e+03	3.91e+02	7.99e+02	1.16e+03	0.00e+00	6.92e+00	1.31e+00	1.18e+00	1.11e+00	0.00e+00	3.70e+00	0.00e+00	1.76e-01	7.24e-01
F112017	2017	10	1.74e+03	5.25e+04	1.05e+04	1.38e+04	1.08e+04	0.00e+00	2.17e+02	2.08e-01	3.42e+01	7.97e+01	0.00e+00	2.17e+02	2.08e-01	1.55e+01	5.35e+01
F122017	2017	10	1.34e+02	1.55e+04	2.45e+03	3.19e+03	2.76e+03	4.09e+00	5.76e+00	4.65e+00	4.62e+00	3.72e-01	0.00e+00	4.36e+00	4.09e+00	4.04e+00	5.80e-01
F132017	2017	10	3.29e+01	2.42e+03	1.20e+02	1.94e+02	3.40e+02	2.58e-01	2.30e+01	2.00e+01	1.41e+01	8.70e+00	0.00e+00	2.02e+01	9.95e-01	3.64e+00	6.36e+00
F142017	2017	10	4.86e+01	4.43e+03	8.09e+02	1.31e+03	1.25e+03	2.66e-04	2.95e+00	3.97e-01	3.82e-01	4.79e-01	6.58e-05	5.00e-01	2.47e-02	1.40e-01	1.94e-01
F152017	2017	10	1.49e+00	8.24e+02	1.29e+02	2.20e+02	2.23e+02	1.95e-04	2.04e+00	3.94e-01	4.21e-01	3.66e-01	1.21e-04	5.00e-01	2.47e-02	1.16e-01	1.59e-01
F162017	2017	10	3.93e+01	2.88e+02	6.37e+01	7.58e+01	3.83e+01	7.53e+00	2.65e+01	2.19e+01	2.05e+01	4.27e+00	2.25e-01	2.07e+01	4.46e+00	7.95e+00	6.67e+00
F172017	2017	10	4.41e+01	5.71e+03	5.20e+02	1.13e+03	1.38e+03	5.85e-04	2.06e+01	4.78e-01	5.43e+00	8.81e+00	1.53e-05	5.00e-01	1.97e-01	2.20e-01	2.12e-01
F182017	2017	10	6.86e+00	2.66e+02	2.25e+01	3.68e+01	4.42e+01	1.17e-01	2.17e+00	6.78e-01	7.55e-01	3.61e-01	5.31e-02	2.62e-01	1.47e-01	1.50e-01	4.43e-02
F192017	2017	10	4.48e+01	1.35e+02	7.43e+01	7.72e+01	2.30e+01	9.25e-01	6.57e+00	2.98e+00	3.13e+00	1.05e+00	1.79e-01	2.05e+00	1.17e+00	1.17e+00	3.88e-01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	1.02e+02	1.14e+02	1.06e+02	1.06e+02	3.10e+00	1.00e+02	1.00e+02	1.00e+02	1.00e+02	1.83e-13	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00
F212017	2017	10	1.00e+02	1.08e+02	1.01e+02	1.02e+02	1.58e+00	1.00e+02	1.01e+02	1.00e+02	1.00e+02	1.95e-01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	1.09e-13
F222017	2017	10	1.04e+02	2.10e+02	1.14e+02	1.18e+02	1.64e+01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	2.17e-13	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00
F232017	2017	10	2.05e+02	2.29e+02	2.13e+02	2.14e+02	6.01e+00	2.00e+02	2.00e+02	2.00e+02	2.00e+02	1.11e-13	2.00e+02	2.00e+02	2.00e+02	2.00e+02	0.00e+00
F242017	2017	10	4.79e+02	4.90e+02	4.81e+02	4.81e+02	1.51e+00	4.76e+02	4.79e+02	4.79e+02	4.79e+02	1.15e+00	4.76e+02	4.79e+02	4.79e+02	4.79e+02	7.03e-01
F252017	2017	10	1.34e+00	4.68e+02	4.67e+02	4.40e+02	1.09e+02	4.55e-13	4.67e+02	4.67e+02	4.39e+02	1.11e+02	0.00e+00	4.67e+02	4.67e+02	4.49e+02	9.16e+01
F262017	2017	10	3.90e+02	4.00e+02	3.92e+02	3.93e+02	2.98e+00	3.90e+02	3.98e+02	3.91e+02	3.90e+02	1.53e+00	3.90e+02	3.91e+02	3.91e+02	3.90e+02	5.06e-01
F272017	2017	10	1.41e+00	2.89e+01	1.55e+01	1.59e+01	7.85e+00	0.00e+00	4.55e-13	4.55e-13	3.48e-13	1.95e-13	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F282017	2017	10	1.09e+03	6.18e+03	1.31e+03	1.57e+03	9.58e+02	9.81e+02	1.06e+03	9.88e+02	1.02e+03	3.28e+01	9.79e+02	1.05e+03	1.04e+03	1.01e+03	3.14e+01
F292017	2017	10	6.62e+03	3.03e+04	1.28e+04	1.37e+04	5.31e+03	1.38e+03	3.16e+03	1.38e+03	1.87e+03	8.00e+02	1.38e+03	3.16e+03	1.38e+03	1.80e+03	7.63e+02
F12022	2022	10	6.27e-02	8.15e+00	8.78e-01	1.80e+00	2.15e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22022	2022	10	5.19e+00	9.07e+00	6.85e+00	7.08e+00	9.45e-01	3.99e+00	8.92e+00	3.99e+00	5.05e+00	2.05e+00	3.99e+00	8.92e+00	3.99e+00	4.47e+00	1.48e+00
F32022	2022	10	2.42e-07	2.49e-05	1.73e-06	3.24e-06	4.18e-06	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F42022	2022	10	1.68e+01	6.60e+01	3.60e+01	3.74e+01	1.18e+01	4.03e+00	1.60e+01	1.00e+01	1.00e+01	2.48e+00	2.00e+00	1.20e+01	8.00e+00	7.80e+00	1.97e+00
F52022	2022	10	2.12e-05	1.25e-02	6.43e-04	1.57e-03	2.23e-03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F62022	2022	10	6.50e+02	2.44e+04	8.29e+03	8.76e+03	5.49e+03	5.26e-02	5.00e-01	4.74e-01	4.12e-01	1.19e-01	8.30e-03	4.98e-01	2.62e-01	2.68e-01	1.51e-01
F72022	2022	10	3.10e+01	7.37e+01	4.45e+01	4.63e+01	9.68e+00	4.40e-01	2.05e+01	1.52e+00	2.33e+00	3.16e+00	4.65e-02	8.40e-01	2.24e-01	2.61e-01	1.90e-01
F82022	2022	10	1.80e+01	3.28e+01	2.90e+01	2.83e+01	3.35e+00	7.52e-01	2.20e+01	1.19e+00	2.91e+00	5.45e+00	9.73e-02	2.02e+01	2.62e-01	6.57e-01	2.79e+00
F92022	2022	10	2.91e-01	3.62e+02	1.44e+00	3.18e+01	9.73e+01	0.00e+00	3.59e+02	4.55e-13	2.11e+01	8.53e+01	0.00e+00	3.59e+02	0.00e+00	2.11e+01	8.53e+01
F102022	2022	10	2.21e+02	2.25e+02	2.23e+02	2.23e+02	1.02e+00	2.21e+02	2.22e+02	2.21e+02	2.21e+02	2.89e-01	2.21e+02	2.22e+02	2.21e+02	2.21e+02	2.55e-01
F112022	2022	10	8.62e-02	6.20e-01	2.40e-01	2.55e-01	1.04e-01	0.00e+00	4.55e-13	4.55e-13	3.92e-13	1.58e-13	0.00e+00	4.55e-13	0.00e+00	1.78e-14	8.91e-14
F122022	2022	10	1.65e+02	1.67e+02	1.66e+02	1.66e+02	4.02e-01	1.64e+02	1.66e+02	1.65e+02	1.65e+02	7.06e-01	1.64e+02	1.66e+02	1.65e+02	1.65e+02	6.64e-01

LSHADE — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	2.92e+08	1.19e+09	6.89e+08	6.78e+08	1.84e+08	4.38e+05	1.91e+06	9.72e+05	9.94e+05	3.05e+05	4.85e+01	9.80e+03	7.68e+02	1.20e+03	1.65e+03
F22014	2014	50	1.08e+10	2.54e+10	1.91e+10	1.86e+10	3.99e+09	1.80e+03	1.86e+04	6.76e+03	7.31e+03	3.16e+03	2.84e-14	5.68e-14	2.84e-14	3.18e-14	9.25e-15
F32014	2014	50	1.21e+05	3.10e+05	2.45e+05	2.38e+05	4.28e+04	4.76e+01	5.66e+02	1.52e+02	1.87e+02	1.08e+02	0.00e+00	5.68e-14	5.68e-14	4.57e-14	2.28e-14
F42014	2014	50	1.48e+03	4.55e+03	2.43e+03	2.56e+03	7.05e+02	9.59e+01	1.04e+02	9.82e+01	9.79e+01	1.26e+00	4.15e+00	9.84e+01	9.81e+01	7.31e+01	4.11e+01
F52014	2014	50	2.11e+01	2.14e+01	2.13e+01	2.13e+01	5.16e-02	2.07e+01	2.12e+01	2.08e+01	2.09e+01	2.07e-01	2.02e+01	2.03e+01	2.03e+01	2.03e+01	3.13e-02
F62014	2014	50	6.39e+01	7.55e+01	7.08e+01	7.06e+01	2.32e+00	6.18e+00	1.55e+01	1.04e+01	1.05e+01	2.25e+00	0.00e+00	1.61e+00	0.00e+00	1.22e-01	3.92e-01
F72014	2014	50	1.38e+02	2.75e+02	1.99e+02	1.99e+02	3.50e+01	4.00e-03	4.56e-02	1.14e-02	1.42e-02	9.16e-03	0.00e+00	1.14e-13	0.00e+00	1.34e-14	3.70e-14
F82014	2014	50	3.82e+02	5.13e+02	4.63e+02	4.61e+02	2.96e+01	9.39e+01	1.50e+02	1.18e+02	1.19e+02	1.18e+01	4.09e-12	8.17e-10	6.08e-11	1.09e-10	1.39e-10
F92014	2014	50	4.63e+02	6.15e+02	5.44e+02	5.41e+02	3.43e+01	1.70e+02	2.59e+02	2.07e+02	2.13e+02	2.17e+01	7.39e+00	1.85e+01	1.23e+01	1.19e+01	2.20e+00
F102014	2014	50	1.15e+04	1.40e+04	1.31e+04	1.31e+04	5.21e+02	3.55e+03	5.51e+03	4.48e+03	4.46e+03	4.44e+02	1.71e-03	1.24e-01	4.59e-02	4.81e-02	2.59e-02
F112014	2014	50	1.34e+04	1.55e+04	1.46e+04	1.46e+04	4.43e+02	7.73e+03	1.21e+04	9.32e+03	9.41e+03	7.04e+02	2.40e+03	4.06e+03	3.22e+03	3.26e+03	3.59e+02
F122014	2014	50	3.44e+00	5.55e+00	4.73e+00	4.62e+00	4.85e-01	8.42e-01	3.83e+00	1.36e+00	1.59e+00	7.45e-01	1.54e-01	2.74e-01	2.10e-01	2.10e-01	2.51e-02
F132014	2014	50	9.84e-01	3.51e+00	2.58e+00	2.48e+00	6.02e-01	2.57e-01	5.52e-01	3.54e-01	3.69e-01	6.79e-02	1.22e-01	2.08e-01	1.55e-01	1.59e-01	1.95e-02
F142014	2014	50	3.30e+01	7.83e+01	5.17e+01	5.29e+01	1.06e+01	3.27e-01	7.23e-01	4.17e-01	4.30e-01	8.74e-02	2.38e-01	5.73e-01	2.97e-01	3.22e-01	6.28e-02
F152014	2014	50	4.70e+03	3.18e+05	5.78e+04	6.72e+04	5.47e+04	1.73e+01	2.86e+01	2.28e+01	2.26e+01	2.37e+00	3.90e+00	6.11e+00	5.07e+00	5.00e+00	5.06e-01
F162014	2014	50	2.24e+01	2.36e+01	2.33e+01	2.32e+01	2.52e-01	2.02e+01	2.29e+01	2.13e+01	2.15e+01	7.24e-01	1.58e+01	1.77e+01	1.69e+01	1.68e+01	4.47e-01
F172014	2014	50	2.79e+07	1.32e+08	6.80e+07	7.16e+07	2.51e+07	1.94e+04	6.96e+04	4.16e+04	4.39e+04	1.23e+04	1.06e+02	1.99e+03	4.02e+02	5.21e+02	3.41e+02
F182014	2014	50	4.64e+08	2.00e+09	1.30e+09	1.25e+09	3.72e+08	5.45e+04	9.26e+04	7.53e+04	7.56e+04	9.06e+03	1.64e+02	2.53e+03	7.05e+02	8.58e+02	5.39e+02
F192014	2014	50	9.90e+06	2.86e+08	9.88e+07	1.14e+08	6.92e+07	3.79e+01	1.22e+02	4.40e+01	4.54e+01	1.14e+01	2.37e+01	3.07e+01	2.64e+01	2.65e+01	1.62e+00
F202014	2014	50	2.97e+10	1.09e+13	1.35e+12	1.82e+12	1.99e+12	2.73e+04	8.83e+04	5.12e+04	5.18e+04	1.29e+04	7.53e+00	4.43e+01	1.70e+01	1.99e+01	8.81e+00
F212014	2014	50	8.15e+07	3.66e+08	1.80e+08	1.84e+08	6.13e+07	8.18e+03	5.75e+04	2.45e+04	2.49e+04	9.71e+03	3.64e+00	2.32e+02	1.07e+01	3.97e+01	7.60e+01
F222014	2014	50	6.86e+08	3.22e+11	3.22e+10	5.93e+10	7.44e+10	9.00e+01	6.69e+02	2.88e+02	2.94e+02	1.57e+02	2.34e+01	2.42e+02	2.41e+01	7.08e+01	9.02e+01
F232014	2014	50	2.87e+02	3.83e+02	3.25e+02	3.26e+02	2.16e+01	2.35e+02	2.35e+02	2.35e+02	2.35e+02	8.46e-05	2.35e+02	2.35e+02	2.35e+02	2.35e+02	2.87e-14
F242014	2014	50	3.70e+02	4.26e+02	3.95e+02	3.95e+02	1.20e+01	2.87e+02	2.96e+02	2.92e+02	2.92e+02	1.70e+00	2.88e+02	2.97e+02	2.91e+02	2.92e+02	1.59e+00
F252014	2014	50	2.73e+02	3.43e+02	3.09e+02	3.11e+02	1.52e+01	2.06e+02	2.08e+02	2.07e+02	2.07e+02	5.99e-01	2.05e+02	2.07e+02	2.05e+02	2.05e+02	3.17e-01
F262014	2014	50	1.04e+02	2.77e+02	1.06e+02	1.19e+02	4.27e+01	1.00e+02	1.01e+02	1.00e+02	1.00e+02	7.05e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	1.82e-02
F272014	2014	50	1.80e+03	2.10e+03	1.96e+03	1.97e+03	6.97e+01	3.72e+02	5.28e+02	4.51e+02	4.48e+02	3.04e+01	3.00e+02	3.63e+02	3.20e+02	3.23e+02	1.85e+01
F282014	2014	50	2.06e+03	6.07e+03	3.07e+03	3.18e+03	6.55e+02	1.09e+03	1.30e+03	1.21e+03	1.21e+03	4.66e+01	1.06e+03	1.19e+03	1.13e+03	1.13e+03	3.04e+01
F292014	2014	50	6.08e+08	1.97e+09	1.02e+09	1.07e+09	2.95e+08	7.33e+03	6.42e+04	1.02e+04	1.16e+04	7.79e+03	4.34e+03	6.50e+03	5.05e+03	5.19e+03	5.49e+02
F302014	2014	50	4.12e+09	3.15e+11	3.94e+10	6.58e+10	7.09e+10	2.69e+05	5.16e+06	8.32e+05	1.05e+06	8.37e+05	1.44e+04	2.86e+04	1.90e+04	1.91e+04	2.45e+03
F12017	2017	50	1.02e+10	2.33e+10	1.73e+10	1.75e+10	3.65e+09	1.20e+03	1.68e+04	5.58e+03	6.08e+03	3.33e+03	1.42e-14	2.84e-14	1.42e-14	1.67e-14	5.47e-15
F22017	2017	50	8.46e+04	1.58e+05	1.22e+05	1.19e+05	1.89e+04	2.98e-01	6.30e+00	1.48e+00	1.78e+00	1.25e+00	0.00e+00	2.84e-14	0.00e+00	2.79e-15	8.54e-15
F32017	2017	50	1.22e+03	4.58e+03	2.36e+03	2.50e+03	7.44e+02	6.94e+01	1.66e+02	1.59e+02	1.57e+02	1.38e+01	5.61e+01	1.56e+02	1.56e+02	1.28e+02	4.37e+01
F42017	2017	50	1.43e+04	2.72e+04	1.88e+04	1.92e+04	3.45e+03	1.69e+02	2.96e+02	2.35e+02	2.33e+02	2.76e+01	6.41e+00	1.60e+01	1.11e+01	1.12e+01	2.17e+00
F52017	2017	50	2.73e-02	5.49e-02	4.31e-02	4.25e-02	7.22e-03	1.81e-06	2.47e-04	4.20e-05	6.32e-05	6.11e-05	1.51e-11	8.57e-10	1.05e-10	1.35e-10	1.23e-10
F62017	2017	50	3.93e+05	1.10e+06	6.98e+05	7.22e+05	1.47e+05	2.67e+02	4.01e+02	3.37e+02	3.36e+02	3.19e+01	5.92e+01	6.90e+01	6.31e+01	6.28e+01	2.06e+00
F72017	2017	50	8.36e+02	1.22e+03	1.07e+03	1.07e+03	6.40e+01	2.58e+02	6.29e+02	3.81e+02	4.03e+02	8.69e+01	3.53e+01	6.41e+01	5.27e+01	5.28e+01	6.26e+00
F82017	2017	50	1.64e+01	4.76e+01	2.85e+01	2.89e+01	6.29e+00	1.14e-05	1.79e-01	9.78e-05	1.25e-02	3.59e-02	0.00e+00	1.14e-13	0.00e+00	2.23e-15	1.59e-14
F92017	2017	50	1.39e+04	1.57e+04	1.49e+04	1.49e+04	5.00e+02	7.38e+03	1.32e+04	9.47e+03	9.64e+03	1.24e+03	2.80e+03	3.94e+03	3.35e+03	3.38e+03	2.60e+02
F102017	2017	50	2.51e+07	5.51e+08	1.43e+08	1.72e+08	1.08e+08	1.74e+03	3.79e+04	5.41e+03	7.68e+03	6.42e+03	1.12e+01	2.50e+02	2.36e+01	3.62e+01	4.16e+01
F112017	2017	50	1.17e+09	3.70e+09	2.31e+09	2.33e+09	6.01e+08	5.41e+04	3.44e+05	1.53e+05	1.68e+05	7.29e+04	7.24e+02	3.13e+03	1.64e+03	1.68e+03	5.58e+02
F122017	2017	50	2.09e+09	7.91e+09	3.54e+09	3.94e+09	1.21e+09	4.03e+04	1.16e+05	8.04e+04	8.13e+04	1.82e+04	2.29e+01	9.33e+02	5.00e+01	1.32e+02	2.00e+02
F132017	2017	50	3.71e+06	2.94e+07	1.15e+07	1.27e+07	5.97e+06	2.05e+03	3.32e+04	1.19e+04	1.32e+04	6.30e+03	2.42e+01	1.34e+02	4.10e+01	5.08e+01	2.69e+01
F142017	2017	50	2.29e+08	1.54e+09	7.81e+08	7.85e+08	2.83e+08	1.38e+04	3.99e+04	2.54e+04	2.64e+04	6.65e+03	2.24e+01	3.92e+02	6.16e+01	1.01e+02	9.47e+01
F152017	2017	50	7.96e+06	1.21e+08	4.11e+07	5.04e+07	3.05e+07	1.77e+01	1.00e+03	2.42e+02	2.81e+02	2.38e+02	4.36e+00	6.56e+02	2.21e+02	1.55e+02	1.60e+02
F162017	2017	50	9.76e+07	9.75e+10	8.25e+09	1.59e+10	2.11e+10	2.60e+02	2.47e+03	1.02e+03	1.05e+03	5.36e+02	2.50e+01	7.23e+02	2.75e+01	4.65e+01	9.94e+01
F172017	2017	50	1.82e+07	9.56e+07	4.14e+07	4.27e+07	1.60e+07	1.31e+04	6.09e+04	3.59e+04	3.72e+04	1.09e+04	2.07e+01	2.84e+01	2.29e+01	2.33e+01	1.60e+00
F182017	2017	50	8.14e+09	2.35e+11	8.01e+10	8.73e+10	4.40e+10	2.49e+03	3.25e+04	1.19e+04	1.42e+04	7.58e+03	5.70e+00	1.84e+01	1.01e+01	1.06e+01	2.96e+00
F192017	2017	50	2.33e+03	5.18e+03	3.45e+03	3.52e+03	6.88e+02	3.74e+01	2.83e+02	5.45e+01	7.86e+01	7.05e+01	2.17e+01	2.41e+02	2.28e+01	3.97e+01	5.90e+01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	6.68e+03	3.30e+04	1.78e+04	1.87e+04	5.14e+03	1.19e+02	5.03e+02	4.30e+02	4.26e+02	6.81e+01	2.07e+02	2.17e+02	2.11e+02	2.11e+02	2.52e+00
F212017	2017	50	4.49e+02	6.96e+02	5.72e+02	5.79e+02	4.86e+01	1.93e+02	2.37e+02	2.21e+02	2.20e+02	8.92e+00	1.00e+02	2.40e+02	2.14e+02	2.12e+02	1.79e+01
F222017	2017	50	1.96e+04	3.30e+04	2.67e+04	2.65e+04	3.01e+03	2.06e+02	2.24e+02	2.12e+02	2.13e+02	4.18e+00	2.00e+02	2.00e+02	2.00e+02	2.00e+02	0.00e+00
F232017	2017	50	1.11e+04	1.64e+04	1.37e+04	1.39e+04	1.10e+03	2.04e+02	2.13e+02	2.07e+02	2.07e+02	2.19e+00	2.00e+02	2.00e+02	2.00e+02	2.00e+02	0.00e+00
F242017	2017	50	2.44e+03	5.18e+03	3.56e+03	3.64e+03	5.29e+02	7.07e+02	7.37e+02	7.26e+02	7.22e+02	1.24e+01	7.06e+02	7.33e+02	7.26e+02	7.19e+02	1.16e+01
F252017	2017	50	2.12e+03	2.83e+03	2.43e+03	2.44e+03	1.58e+02	1.27e+03	1.63e+03	1.62e+03	1.54e+03	1.12e+02	1.18e+03	1.63e+03	1.62e+03	1.48e+03	2.00e+02
F262017	2017	50	8.72e+02	1.33e+03	1.04e+03	1.04e+03	1.06e+02	4.88e+02	5.44e+02	5.08e+02	5.09e+02	9.64e+00	4.81e+02	5.22e+02	5.00e+02	5.00e+02	9.33e+00
F272017	2017	50	7.37e+02	1.05e+03	8.62e+02	8.83e+02	9.50e+01	3.84e+01	4.62e+02	6.55e+01	9.21e+01	7.45e+01	4.55e-13	9.09e-13	4.55e-13	5.53e-13	1.89e-13
F282017	2017	50	3.15e+09	7.47e+11	5.43e+10	8.10e+10	1.11e+11	8.77e+03	4.59e+04	2.44e+04	2.52e+04	9.59e+03	2.82e+03	6.53e+03	3.79e+03	4.01e+03	8.90e+02
F292017	2017	50	1.61e+10	1.65e+11	4.14e+10	4.96e+10	3.15e+10	1.11e+05	6.17e+05	2.17e+05	2.48e+05	1.03e+05	6.06e+03	2.43e+04	9.20e+03	1.08e+04	4.57e+03
F12022	2022	20	6.47e+02	4.62e+03	1.95e+03	1.96e+03	8.85e+02	5.68e-14	2.84e-13	5.68e-14	7.69e-14	4.66e-14	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22022	2022	20	5.42e+01	1.05e+02	7.06e+01	7.13e+01	9.82e+00	4.49e+01	4.91e+01	4.91e+01	4.88e+01	9.95e-01	4.49e+01	4.91e+01	4.91e+01	4.87e+01	1.26e+00
F32022	2022	20	1.73e-03	1.09e-02	4.40e-03	4.93e-03	2.28e-03	0.00e+00	1.14e-13	0.00e+00	4.46e-15	2.23e-14	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F42022	2022	20	1.56e+02	2.91e+02	2.33e+02	2.31e+02	3.25e+01	2.05e+01	7.07e+01	3.98e+01	4.14e+01	1.12e+01	1.00e+01	2.80e+01	1.80e+01	1.89e+01	3.86e+00
F52022	2022	20	2.83e-01	1.66e+00	5.88e-01	6.19e-01	2.32e-01	0.00e+00	1.14e-13	1.14e-13	7.36e-14	5.49e-14	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F62022	2022	20	6.38e+05	3.12e+07	3.67e+06	5.67e+06	5.73e+06	8.84e+00	3.34e+03	1.50e+02	3.17e+02	5.34e+02	3.90e-01	5.00e-01	4.98e-01	4.86e-01	2.46e-02
F72022	2022	20	9.36e+01	5.84e+02	2.41e+02	2.60e+02	1.13e+02	2.22e+01	3.24e+01	2.51e+01	2.56e+01	1.98e+00	1.61e+01	2.12e+01	2.06e+01	2.04e+01	1.01e+00
F82022	2022	20	3.20e+02	4.48e+03	1.65e+03	1.74e+03	8.98e+02	5.18e+00	2.84e+01	2.47e+01	2.24e+01	6.33e+00	7.77e-01	2.15e+01	2.14e+00	1.00e+01	9.63e+00
F92022	2022	20	3.52e+02	4.11e+02	3.67e+02	3.68e+02	1.18e+01	3.36e+02	3.36e+02	3.36e+02	3.36e+02	3.43e-12	3.36e+02	3.36e+02	3.36e+02	3.36e+02	2.15e-13
F102022	2022	20	4.21e+02	5.50e+02	4.37e+02	4.44e+02	2.43e+01	3.95e+02	4.03e+02	3.99e+02	3.99e+02	1.93e+00	3.97e+02	4.01e+02	3.97e+02	3.98e+02	1.57e+00
F112022	2022	20	7.53e+00	2.49e+01	1.76e+01	1.74e+01	3.75e+00	2.55e-11	7.17e-10	1.26e-10	1.67e-10	1.36e-10	4.55e-13	4.55e-13	4.55e-13	4.55e-13	0.00e+00
F122022	2022	20	2.47e+02	2.77e+02	2.54e+02	2.54e+02	4.58e+00	2.38e+02	2.45e+02	2.38e+02	2.39e+02	1.42e+00	2.38e+02	2.39e+02	2.38e+02	2.38e+02	4.04e-01